

Lesson 3

Data Cleaning & Transformation with Power Query Editor

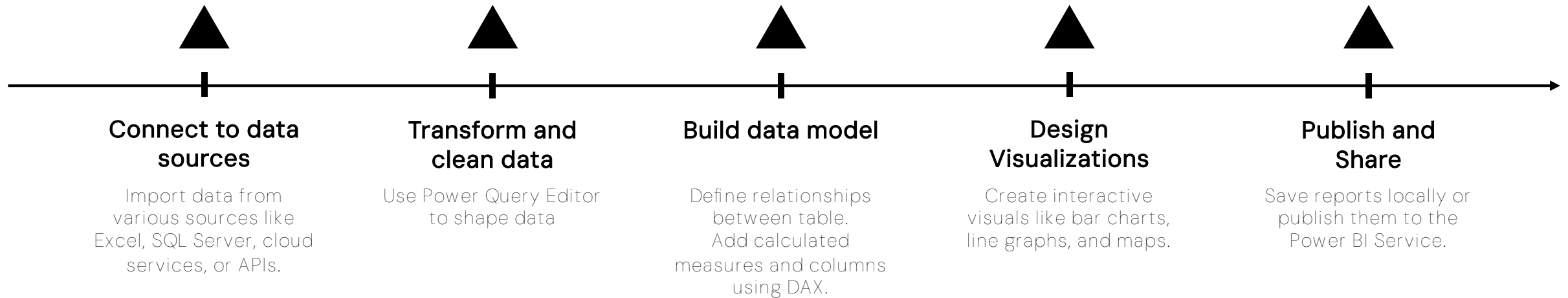
Prepare & Optimize Your Data for Analysis

March 2025

Power BI Desktop workflow.

From Data to Insights

Workflow



Connect → Transform → Model → Visualize → Publish

Power BI Desktop streamlines the process from raw data to actionable insights, enabling better decision-making.

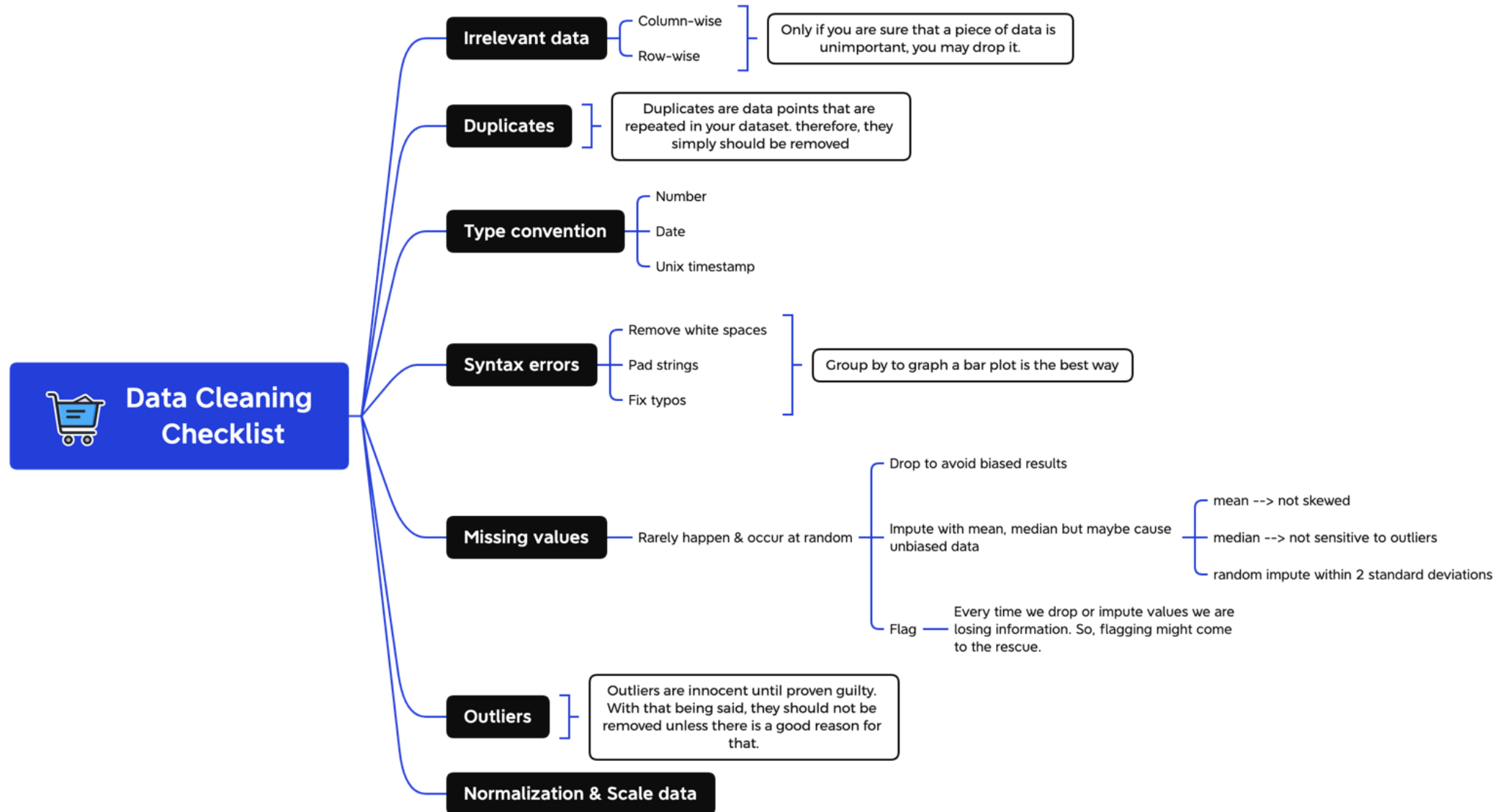
Power BI Desktop **workflow.**

From Data to Insights

Before cleaning data ...

Before applying data cleaning techniques, students need to **understand the structure, quality, and characteristics of their dataset.** Data profiling in Power Query Editor helps identify data issues, inconsistencies, and anomalies before performing transformations.

Data cleaning.



What is Data Profiling?

Definition

Data profiling is the process of examining data to understand its **structure, quality, and distribution**. It helps identify anomalies, patterns, and potential data quality issues.

▷ 3 Key Data Profiling Features in Power Query:



Column Quality:

Detecting Data Issues



Column Distribution:

Identifying Data Patterns
& Outliers



Column Profile:

Detailed Statistical Insights

Data profiling: Column Quality

Definition

Displays a summary of data quality for each column using visual indicators.

→ Quickly identify and address errors or missing data

customer_id	
Valid	100%
Error	0%
Empty	0%
	1
	2

Column quality shows the percentage of values within a column that are **valid**, **contain errors**, or are **empty**

Hover over the column quality box to see the **number of records** in each category

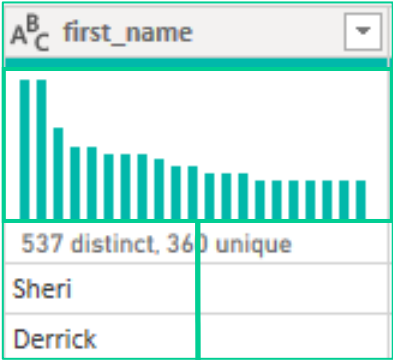
first_name	last_name	customer_id
Valid	Valid	Valid
Error	Error	Error
Empty	Empty	Empty
1000 (100%)	0 (0%)	0 (0%)
Valid	Error	Empty
Rebecca	Kar	2433 Bai
Kim	Br	2219 De
Brenda	Blu	7640 Fir
Darren	Sta	337 Tos
Jonathan	Mu	8668 Via
Jewel	Cre	
Peggy	Me	
Bryan	Rut	

Click the **options menu** to remove duplicates, errors or empty values

Data profiling: Column Distribution

Definition

Provides a visual representation of the frequency and uniqueness of values in a column.
→ **Identify patterns, duplicates, and anomalies in the data.**

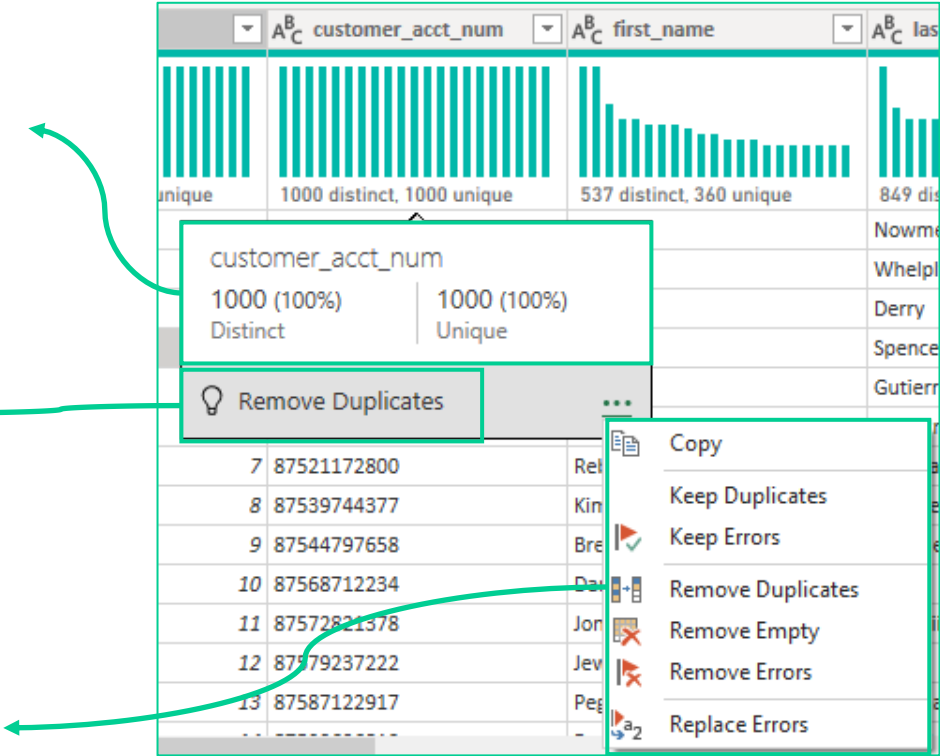


Column distribution
provides a sample
distribution of the data in a
column

Hover over the column quality
box to see the **number of
distinct & unique records**

Suggested action based
on column distribution
results

Click the **options menu** to
remove duplicates, errors
or empty values



Data profiling: Column Profile

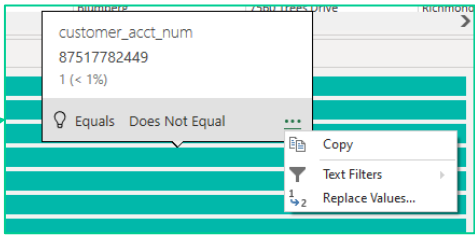
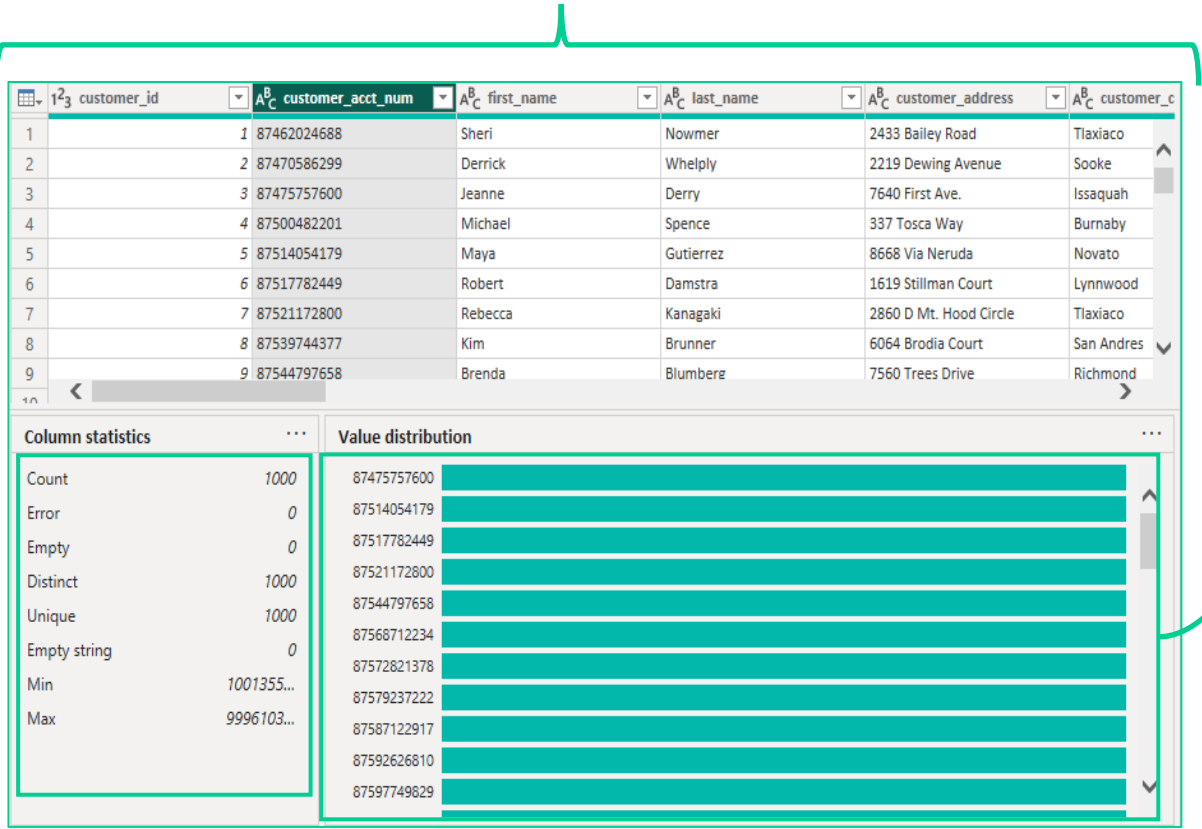
Definition

Offers a detailed statistical overview of a selected column.

→ **Deep dive into individual column properties to understand its structure and distribution.**

Column profile provides a more holistic view of the data in a column, including a sample distribution and profiling statistics

Column statistics
provide more detailed
profiling metrics



Hover over the value distribution
bar for **suggested transformations**
and additional options

What are Text tools?

Definition

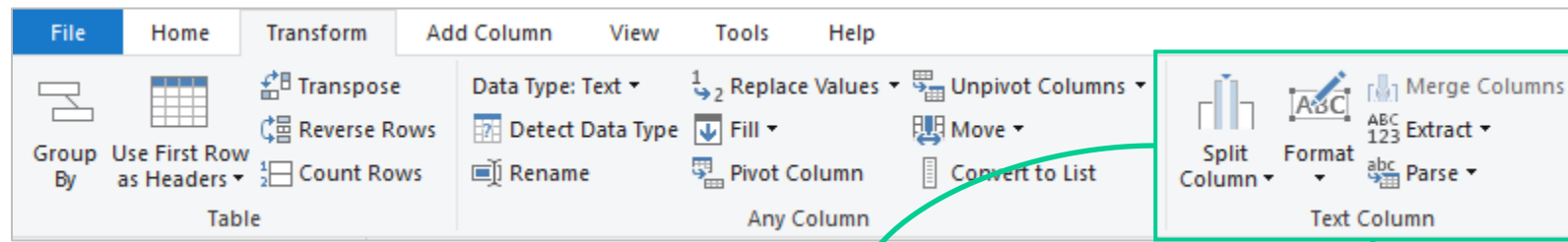
Power BI provides a variety of text tools for transforming and managing text data. These tools allow users to clean, format, and analyze text columns efficiently.

Text Tools: Split Column

Definition

Splits a text column into multiple columns based on a specified delimiter or fixed width.

→ **Break down composite text values (e.g., full names into first and last names).**



Split a text column based on a specific delimiter, number of characters, or other attributes

Format a text column to upper, lower or proper case, or add a prefix or suffix

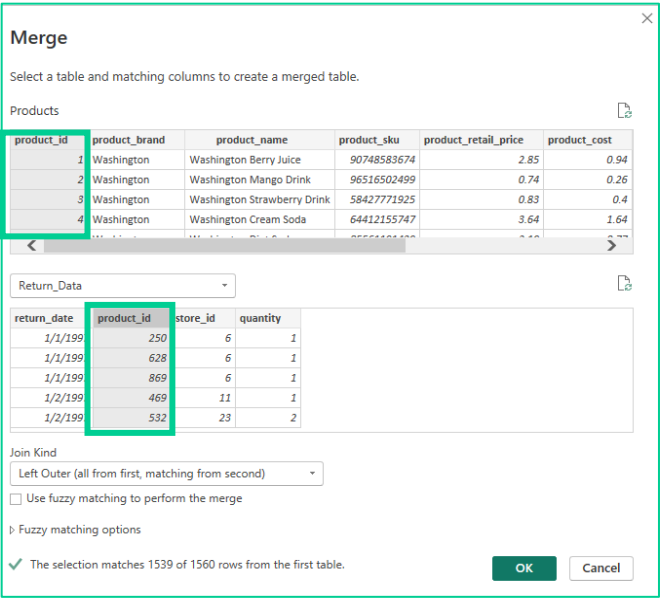
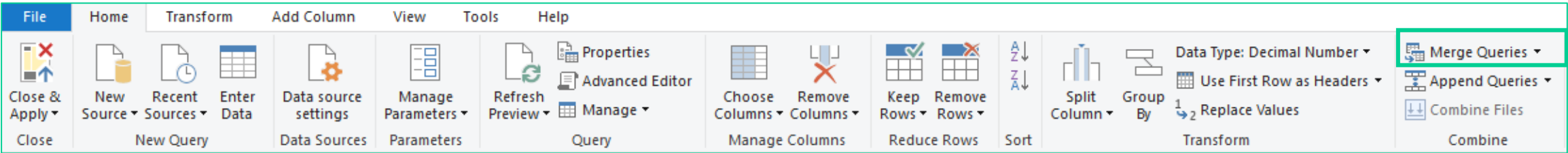
Extract characters from text based on fixed lengths, first/last characters, ranges or delimiters

Text Tools: Merge Columns

Definition

Combines multiple columns into a single text column..

→ **Concatenate columns to create a unified text field, such as combining first and last names.**



Merging queries allows you to join tables based on a common column (like a lookup in Excel)

Text Tools: Replace Values

Definition

..

→ ...

Text Tools: Replace Errors

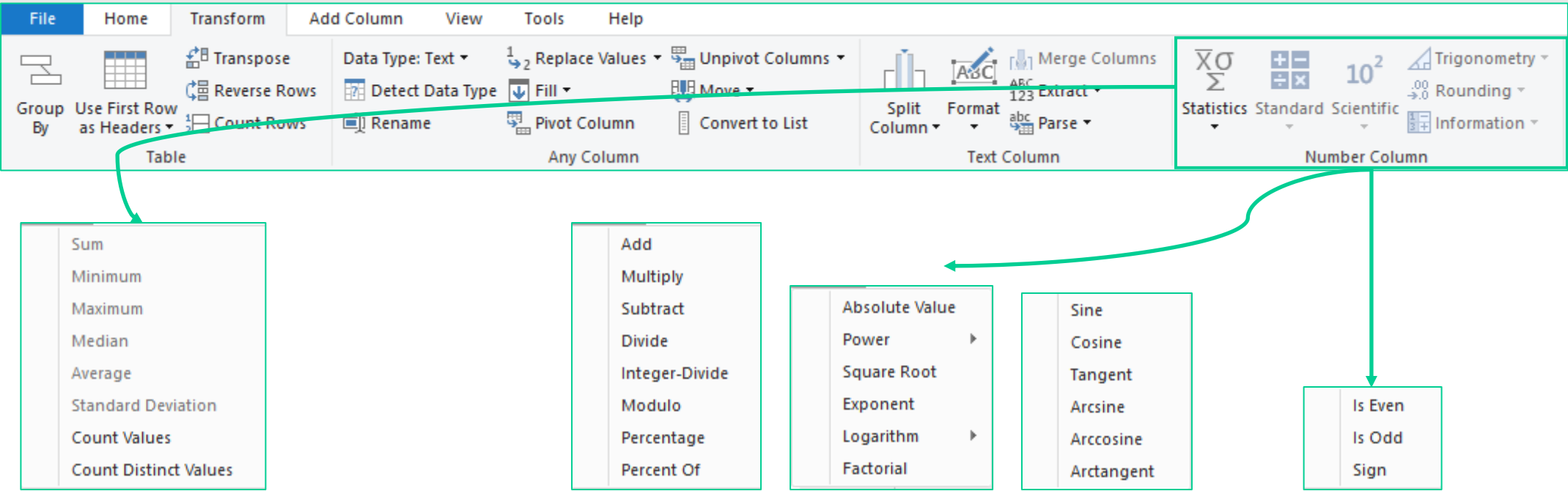
Definition

..
→ ...

What are Numerical tools?

Definition

These tools are essential for data cleaning, transformation, analysis, and visualization..



Statistics functions allow you to evaluate basic stats for a selected column (sum, min/max, average, count, count distinct, etc.)

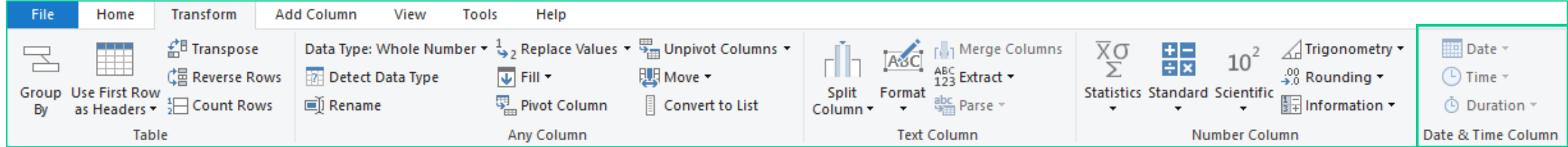
Standard, Scientific and Trigonometry tools allow you to apply standard operations (addition, multiplication, division, etc.) or more advanced calculations (power, logarithm, sine, tangent, etc.) to each value in a column

Information tools allow you to define binary flags (1/0 or TRUE/FALSE) to mark rows as even, odd, positive or negative

Date & Time Tools

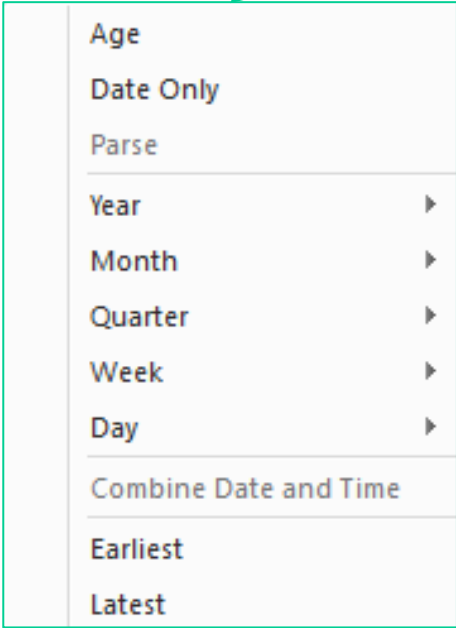
Definition

Powerful tools for managing, transforming, and analyzing date and time data



Date & Time tools are relatively straight-forward, and include the following options:

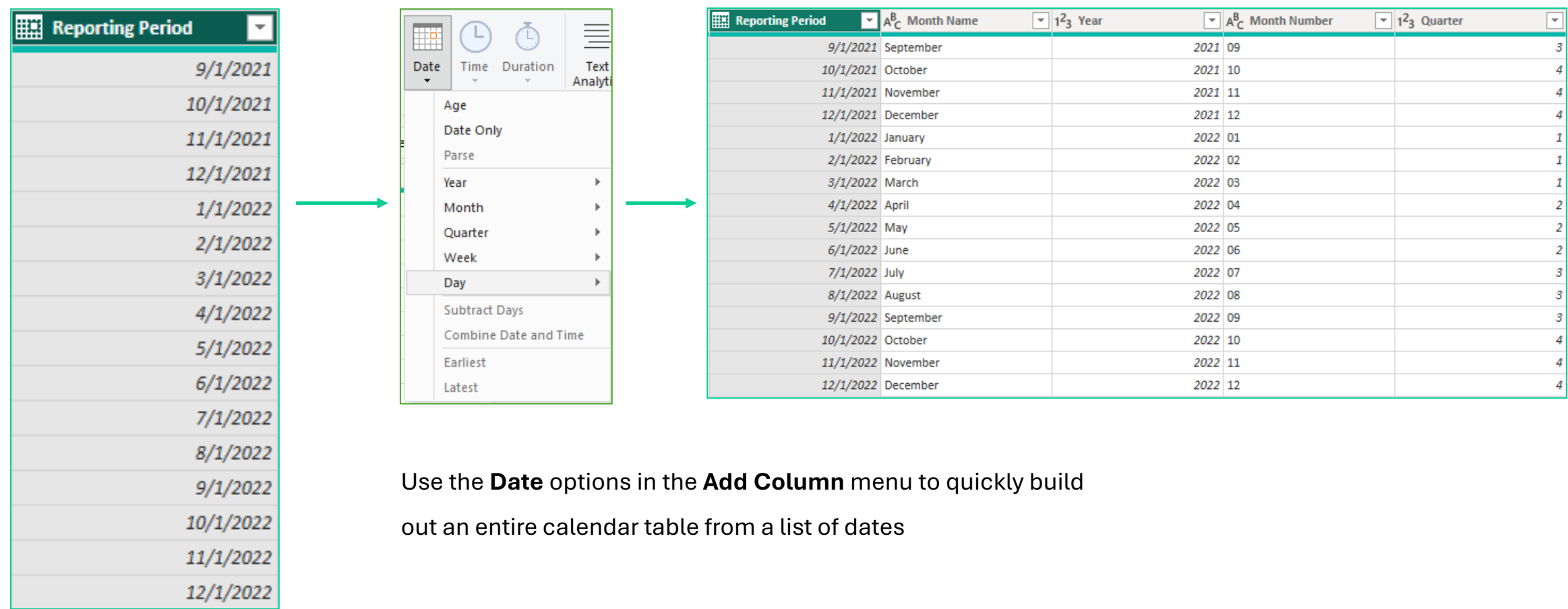
- **Age:** Difference between the current date and the date in each row
- **Date Only:** Removes the time component from a date/time field
- **Year/Month/Quarter/Week/Day:** Extracts individual components from a date field (timespecific options include Hour, Minute, Second, etc.)
- **Earliest/Latest:** Evaluates the earliest or latest date from a column as a single value (can only be accessed from the “Transform” menu)



Creating a calendar table.

Definition

Creating a Calendar Table in Power BI is essential for robust date and time analysis, especially when you need to create relationships with other tables or perform time intelligence calculations.

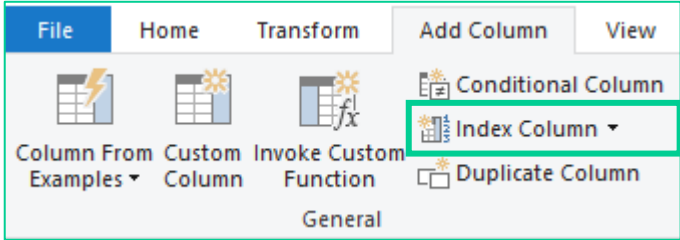


Use the **Date** options in the **Add Column** menu to quickly build out an entire calendar table from a list of dates

Index columns

Definition

A column that contains sequential numbers for each row in a dataset. It can be used for tasks such as creating unique identifiers, ordering data, or referencing rows for calculations



Index Columns contain a list of sequential values that can be used to identify each unique row in a table (typically starting from 0 or 1)

These are often used to create **unique IDs** that can be used to form relationships between tables (more on that later!)

1 ² 3 Index	1 ² 3 customer_id	A ^B _C customer_acct_num	A ^B _C first_name	A ^B _C last_name	A ^B _C customer_a
1		1 87462024688	Sheri	Nowmer	2433 Bailey F
2		2 87470586299	Derrick	Whelply	2219 Dewing
3		3 87475757600	Jeanne	Derry	7640 First Av
4		4 87500482201	Michael	Spence	337 Tosca W
5		5 87514054179	Maya	Gutierrez	8668 Via Ner
6		6 87517782449	Robert	Damstra	1619 Stillma
7		7 87521172800	Rebecca	Kanagaki	2860 D Mt. H
8		8 87539744377	Kim	Brunner	6064 Brodia
9		9 87544797658	Brenda	Blumberg	7560 Trees D
10		10 87568712234	Darren	Stanz	1019 Kenwal
11		11 87572821378	Jonathan	Murraiin	5423 Camby
12		12 87579237222	Jewel	Creek	1792 Belmor
13		13 87587122917	Peggy	Medina	3796 Keller F
14		14 87592626810	Bryan	Rutledge	3074 Ardith I
15		15 87597749829	Walter	Cavestany	7987 Seawin
16		16 87603285908	Peggy	Planck	4864 San Car
17		17 87625473141	Brenda	Marshall	2687 Ridge R
18		18 87637655735	Daniel	Wolter	2473 Orchar
19		19 87650814652	Dianne	Collins	551 Rainier C
20		20 87653979700	Beverly	Baker	591 Merriew
21		21 87663244009	Pedro	Castillo	1579 Plaza R
22		22 87675641200	Laurie	Borges	1873 Lyon Ci
23		23 87678398489	Shauna	Wyro	3114 Notre C

Add Conditional Columns

Definition

Conditional Columns allow you to define new fields based on logical rules and conditions (**IF/THEN statements**)

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Custom

	Column Name	Operator	Value		Output
If			ABC 123		
				Then	ABC 123

Add Clause

Else

ABC 123

OK Cancel

Conditional Columns allow you to define new fields based on logical rules and conditions (IF/THEN statements)

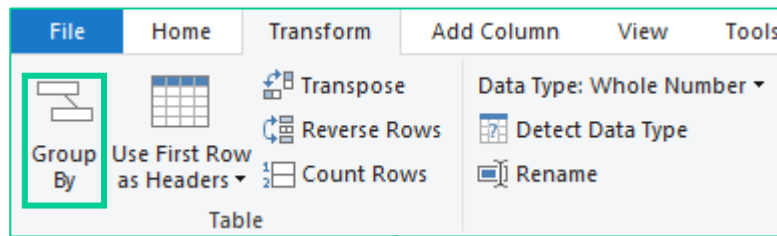
Here we're creating a conditional column named **Quantity Type**, which is based on **Order Quantity**:

- If Order Quantity =1, Quantity Type = “**Single Item**”
- Else If Order Quantity >1, Quantity Type = “**Multiple Items**”
- Else; Quantity Type = “**Other**”

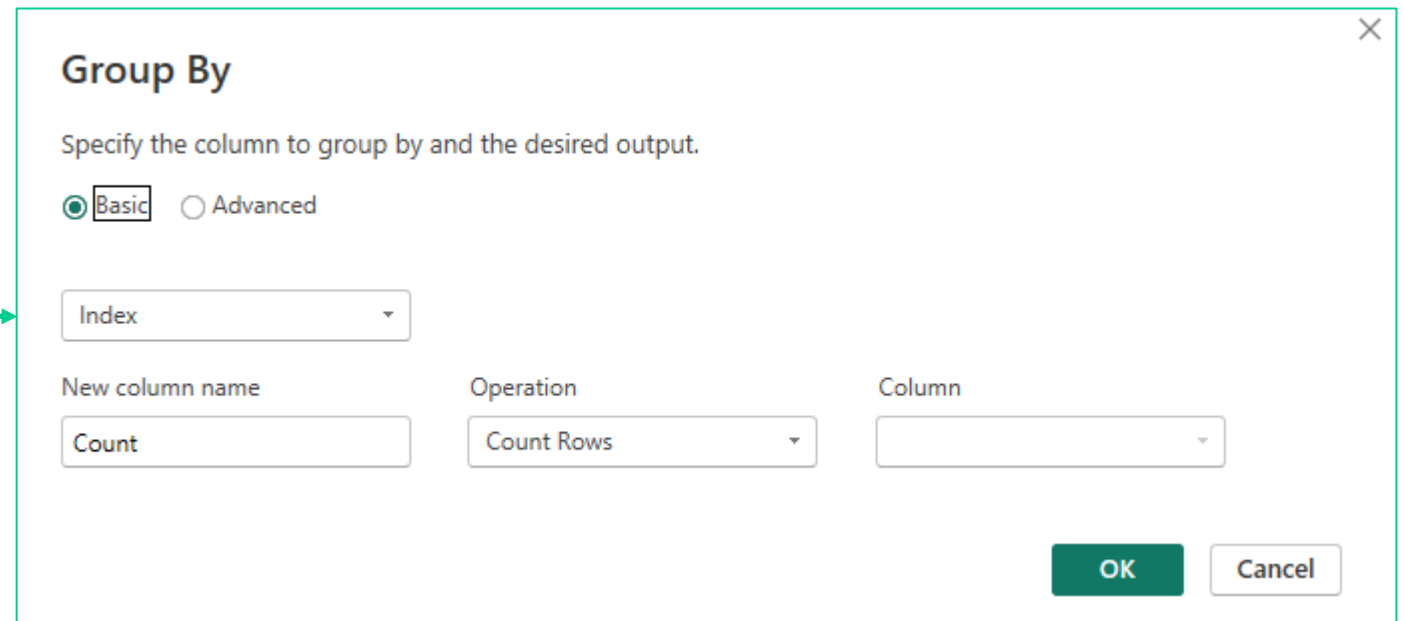
Grouping & Aggregating

Definition

Groups rows based on a text column and performs aggregation operations (e.g., count, concatenate).



Group By allows you to aggregate data at a different level or “grain” (i.e. group daily records into monthly, aggregate transactions by store, etc.)



Pivoting & Unpivoting.

Definition

Pivoting in Power BI involves transforming rows into columns, typically to summarize data.

Unpivoting in Power BI transforms columns into rows, often used to normalize data into a long format.

	A ^B C product_name	1.2 product_sku	1.2 product_retail_price	1.2 product_cost	1.2 product_weight
1	Washington Berry Juice	90748583674	2.85	0.94	8.39
2	Washington Mango Drink	96516502499	0.74	0.26	7.42
3	Washington Strawberry Drink	58427771925	0.83	0.4	13.1
4	Washington Cream Soda	64412155747	3.64	1.64	10.6
5	Washington Diet Soda	85561191439	2.19	0.77	6.66
6	Washington Cola	29804642796	1.15	0.37	15.8
7	Washington Diet Cola	20191444752	2.61	0.91	18
8	Washington Orange Juice	89770532254	2.59	0.8	8.97
9	Washington Cranberry Juice	49395100474	2.42	0.77	7.14
10	Washington Apple Juice	22114084366	1.42	0.5	8.13
11	Washington Apple Drink	17074288722	3.51	1.65	20
12	Jeffers Oatmeal	49031038884	1.54	0.69	8.9
13	Jeffers Corn Puffs	13229009505	2.65	1.27	10.4
14	Jeffers Wheat Puffs	92942813034	1.93	0.71	21.6
15	Jeffers Grits	26378549933	2.29	1.08	21.3
16	Blue Label Canned Beets	62908702499	3.83	1.72	21.2
17	Blue Label Creamed Corn	79484335784	2.99	1.5	6.91
18	Blue Label Canned String Beans	85252254603	2.67	1.17	12.6
19	Blue Label Chicken Soup	47163524033	3.19	1.47	15.2
20	Blue Label Canned Yams	22169209122	2.78	0.92	21.3
21	Blue Label Vegetable Soup	43318244814	2.33	0.89	19.7
22	Blue Label Canned Tomatos	77561696177	2.29	0.98	15
23	Blue Label Noodle Soup	32829326984	1.75	0.65	10.6
24	<				

Pivoting describes the process of turning **distinct row values into columns**, and **unpivoting** describes the process of turning **distinct columns into rows**

UNPIVOT

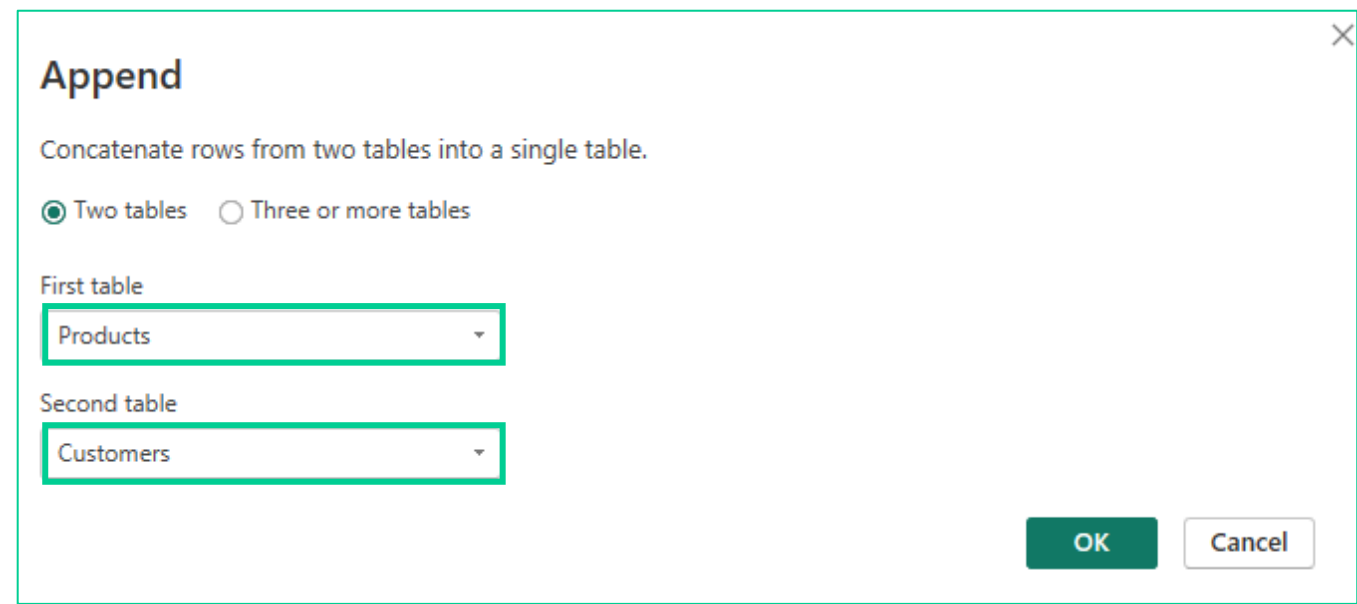
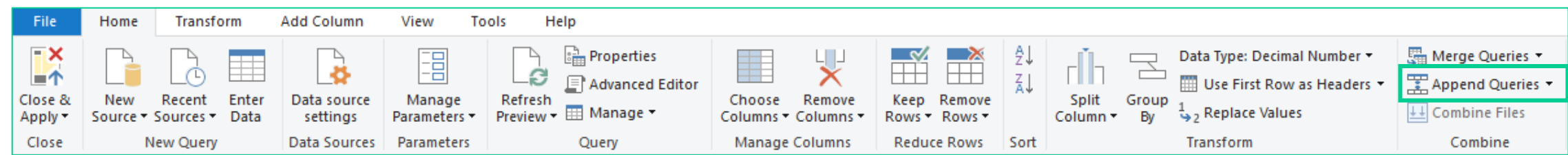
	A ^B C product_name	1.2 product_sku	A ^B C Attribute	1.2 Value	1.2 recyclable	1.2 lo
1	Washington Berry Juice	90748583674	product_retail_price	2.85	0	
2	Washington Berry Juice	90748583674	product_cost	0.94	0	
3	Washington Berry Juice	90748583674	product_weight	8.39	0	
4	Washington Mango Drink	96516502499	product_retail_price	0.74	0	
5	Washington Mango Drink	96516502499	product_cost	0.26	0	
6	Washington Mango Drink	96516502499	product_weight	7.42	0	
7	Washington Strawberry Drink	58427771925	product_retail_price	0.83	1	
8	Washington Strawberry Drink	58427771925	product_cost	0.4	1	
9	Washington Strawberry Drink	58427771925	product_weight	13.1	1	
10	Washington Cream Soda	64412155747	product_retail_price	3.64	1	
11	Washington Cream Soda	64412155747	product_cost	1.64	1	
12	Washington Cream Soda	64412155747	product_weight	10.6	1	
13	Washington Diet Soda	85561191439	product_retail_price	2.19	1	
14	Washington Diet Soda	85561191439	product_cost	0.77	1	
15	Washington Diet Soda	85561191439	product_weight	6.66	1	
16	Washington Cola	29804642796	product_retail_price	1.15	0	
17	Washington Cola	29804642796	product_cost	0.37	0	
18	<					

PIVOT

Pivoting & Unpivoting.

Definition

Appending queries in Power BI combines rows from two or more tables (or queries) into a single table. It's useful when consolidating datasets with the same structure, such as monthly reports or data from different sources.

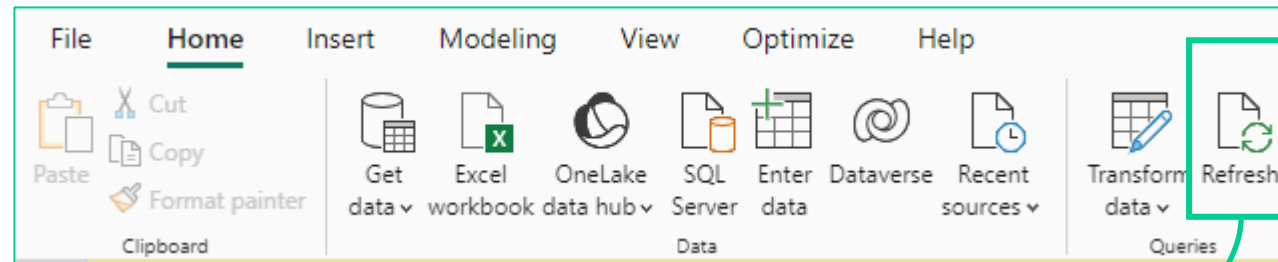


Appending queries allows you to combine or stack tables sharing the exact same column structure and data types

Refreshing queries.

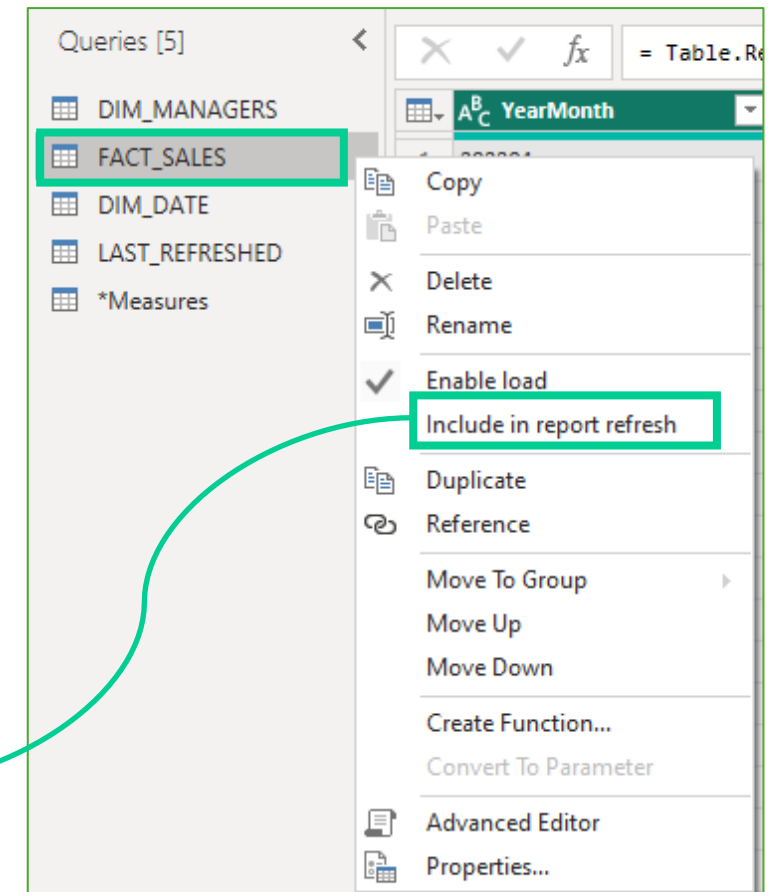
Definition

Appending queries in Power BI combines rows from two or more tables (or queries) into a single table. It's useful when consolidating datasets with the same structure, such as monthly reports or data from different sources.



By default, **all queries** will refresh when you use the **Refresh** command from the **Home** tab

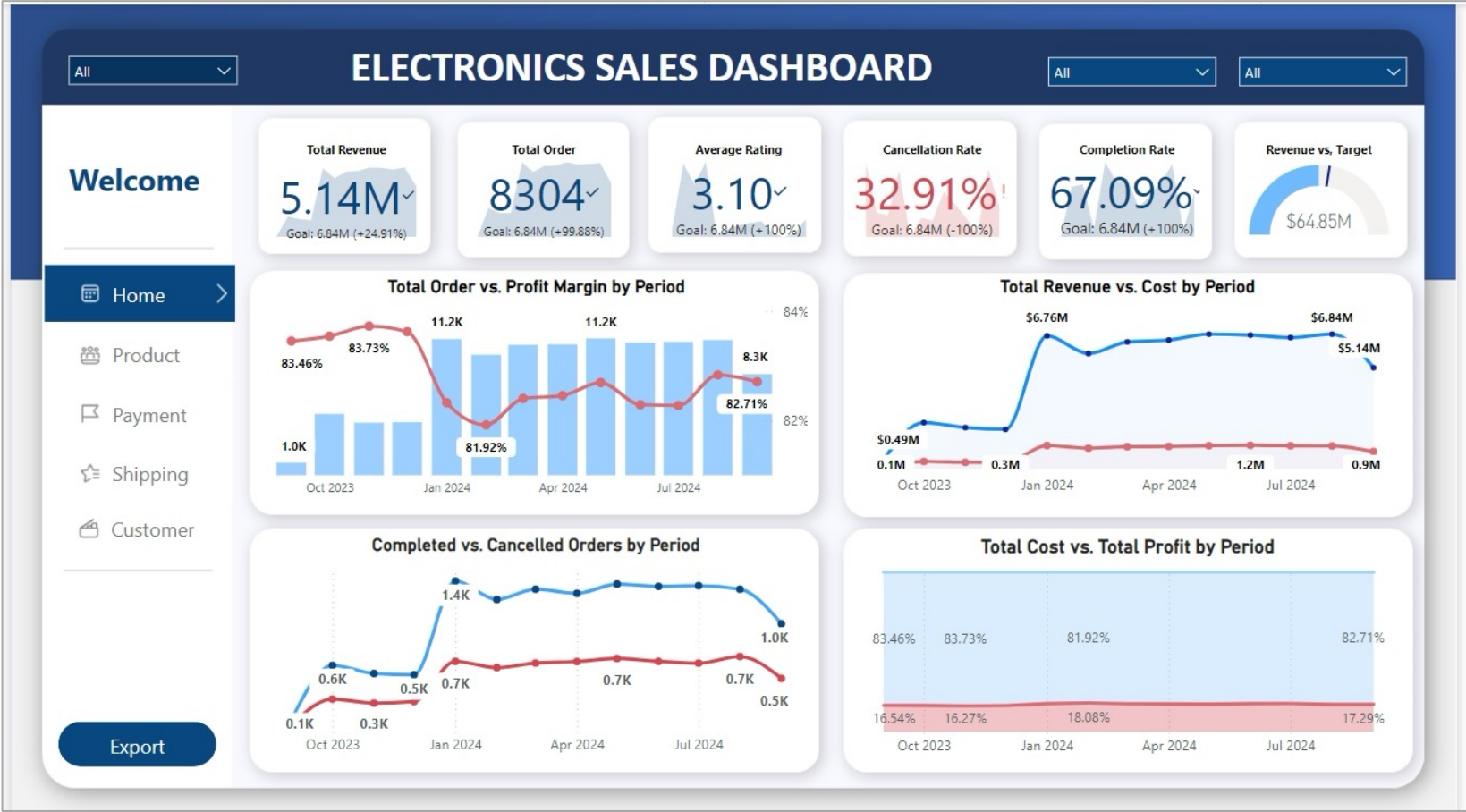
From the **Query Editor**, uncheck **Include in report refresh** to exclude individual queries from the refresh



Dashboard Layout & Structure.

A structured layout is crucial for a user-friendly dashboard

Header section



Navigation Panel

Sections for navigation

Slicers

Allow users to filter by date, product, or location

Main visualization area

Contains various charts and KPI cards.